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Identification of keystone taxa shaping biocrust formation and biodeterioration of limestone monuments in the Xiaoling Tomb of the Ming Dynasty

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Abstract

The limestone monuments of the Rectangular Tower in the Xiaoling Tomb of the Ming Dynasty, created in the mid-fourteenth century, are biodeteriorating from environmental exposure, resulting in the formation of black biocrusts. However, the microbiomes that shape biocrust formation and the biodeterioration processes involved remain unclear, significantly challenging the conservation of stone monuments at this archaeological site. Here, we systematically investigated the physicochemical properties and microbial communities of biocrusts to identify keystone taxa that shape their formation and biodeterioration. Physicochemical analysis indicated that biological crusts are associated with calcium mobilization and redistribution of the limestone monuments. Microscopy and spectroscopy indicated that microbial interactions with limestone promote the formation of biological crusts. Importantly, we observed the significant predominance of Cyanobacteria and/or Chloroflexi in biocrusts, suggesting that photosynthesis may be a crucial process in biocrust formation. Fungal communities in biocrusts were dominated by Ascomycota, Basidiomycota, and Chytridiomycota, while archaeal communities were dominated solely by Nitrososphaerota. Microbial co-occurrence network and correlation analyses identified 12 keystone taxa across 11 genera that shape biocrust formation. Importantly, *Scytonema* spp. could provide organic carbon and nitrogen for *Spirosomaceae* spp., and members of the classes Cyanobacteriia and Agaricomycetes, as well as the genera *Setophaeosphaeria* and *Plectosphaerella*, are likely the keystone taxa responsible for both biocrust formation and the associated biodeterioration. Additionally, two predominant ammonia-oxidizing archaeal families (i.e., *Nitrososphaeraceae* and *Candidatus Nitrocosmicus*) could support chemolithoautotrophic growth in the microbiome by oxidizing ammonia and fixing carbon dioxide. Together, these findings underscore the need for targeted conservation strategies to mitigate microbial biodeterioration of stone monuments during biocrust formation.

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Introduction

Stone monuments serve as crucial carriers of human civilization and are widely distributed worldwide. They not only embody the outstanding achievements of ancient art and craftsmanship but also stand as tangible witnesses to historical transformations. However, during long-term open-air preservation, these heritage sites are generally subjected to dual biotic and abiotic damage, leading to various forms of deterioration, including cracks, exfoliation, salt crystallization, and microbial colonization. Among these, biodeterioration caused by microbial colonization, growth, and metabolism poses a severe challenge to the long-term conservation of stone monuments [1, 2].

Constructed in the mid-fourteenth century, the Imperial Xiaoling Mausoleum is a representative example of Ming and Qing imperial tombs and was inscribed on the UNESCO World Heritage List in 2003. As part of the Imperial Xiaoling Mausoleum, the Rectangular Tower has long been exposed to a warm, humid subtropical climate. Its surfaces have developed prominent biological erosion characterized by the formation of black biocrusts [3]. Such biocrusts not only extensively cover the walls, significantly impairing the heritage's aesthetic appearance, but may also accelerate mineral dissolution and structural deterioration, posing a potential threat to the long-term preservation of the site.

Biocrusts typically consist of a complex microecosystem formed by various microorganisms—including bacteria, fungi, algae, and archaea—along with mineral particles and extracellular polymeric substances [4, 5]. Photoautotrophic organisms, such as algae and cyanobacteria, are key contributors to biocrust formation [6]. They can colonize outdoor stone monuments with only water present and are often regarded as pioneers of microbial colonization of stone surfaces [7]. Heterotrophic organisms, in turn, consume the carbon compounds released by photoautotrophs [8]. During mutual metabolism, microorganisms may secrete acidic substances that corrode building surfaces, while others deposit pigments that lead to discoloration [9, 10]. Fungal hyphae can penetrate the brick and stone, causing physical damage. Moreover, the high water-retention capacity of biocrusts reduces evaporation rates, thereby creating a favorable microenvironment for microbial survival and accelerating biodeterioration [11].

Previous studies have shown that environmental factors significantly influence the microbial community structure of brick- and stone-based heritage. For instance, the bacterial and fungal communities on the surfaces of the Florence Cathedral are affected by temperature and humidity [12]. The diversity of fungal communities in subterranean tomb murals is influenced by temperature and relative humidity [13]. However, a systematic understanding of

the microbial composition of biocrusts on stone monuments and the key factors shaping biocrust formation remains limited. In particular, the synergistic mechanisms among microorganisms from the three domains of life (bacteria, fungi, and archaea) in biocrust formation and biodeterioration remain poorly understood.

Therefore, to gain a more comprehensive and in-depth understanding of the microbiological mechanisms underlying biocrust formation and the biodeterioration of stone monuments, this study focuses on the typical black biocrusts on the limestone walls of the Rectangular Tower in the Imperial Xiaoling Mausoleum. Using high-throughput sequencing, we simultaneously analyzed the community structures of bacteria, fungi, and archaea in the biocrusts. By constructing microbial co-occurrence networks and analyzing topological attributes and multivariate statistics, we aimed to identify keystone species that play critical roles in biocrust formation and biodeterioration. Furthermore, we seek to elucidate their potential functions in the biodeterioration of stone monuments by correlating these key taxa with physicochemical factors. This research aims to provide a scientific basis for understanding the ecological mechanisms underlying biocrust-induced deterioration and to develop targeted conservation strategies for stone cultural heritage.

Materials and methods

Sample collection

Samples were collected in May 2025 in Nanjing (118° 49' 22" E, 32° 33' 51" N), Jiangsu Province, China. This study employed a non-destructive method to collect the widely distributed black biocrust samples from the limestone wall of the Rectangular Tower in the Xiaoling Tomb of the Ming Dynasty. Approximately 5 g of crust samples were gently collected with sterile tweezers into sterile sampling bags, without damaging the monument materials. Specifically, three samples (C1, C2, and C3) were collected from the right corner of the Rectangular Tower, and the other three (C4, C5, and C6) were collected from the left corner (Fig. 1a). All six samples were used as biological replicates for the study. All samples were packaged and stored at 4 °C before subsequent analyses.

Ion chromatography analysis

The biocrust samples were ground into a homogeneous powder using a quartz mortar (Fig. 1b). Next, 1 g of the powder was added to 7 mL of ultrapure water, mixed for 15 min in a vortex mixer, and then extracted in an ultrasonic bath for 10 min to obtain a soluble salt solution. Subsequently, the solution was centrifuged at 5000 g for 5 min to collect the supernatant, which was then filtered through a 0.22 µm microporous membrane for subsequent determination of anion and cation concentrations. To meet the ion chromatography detection limit,

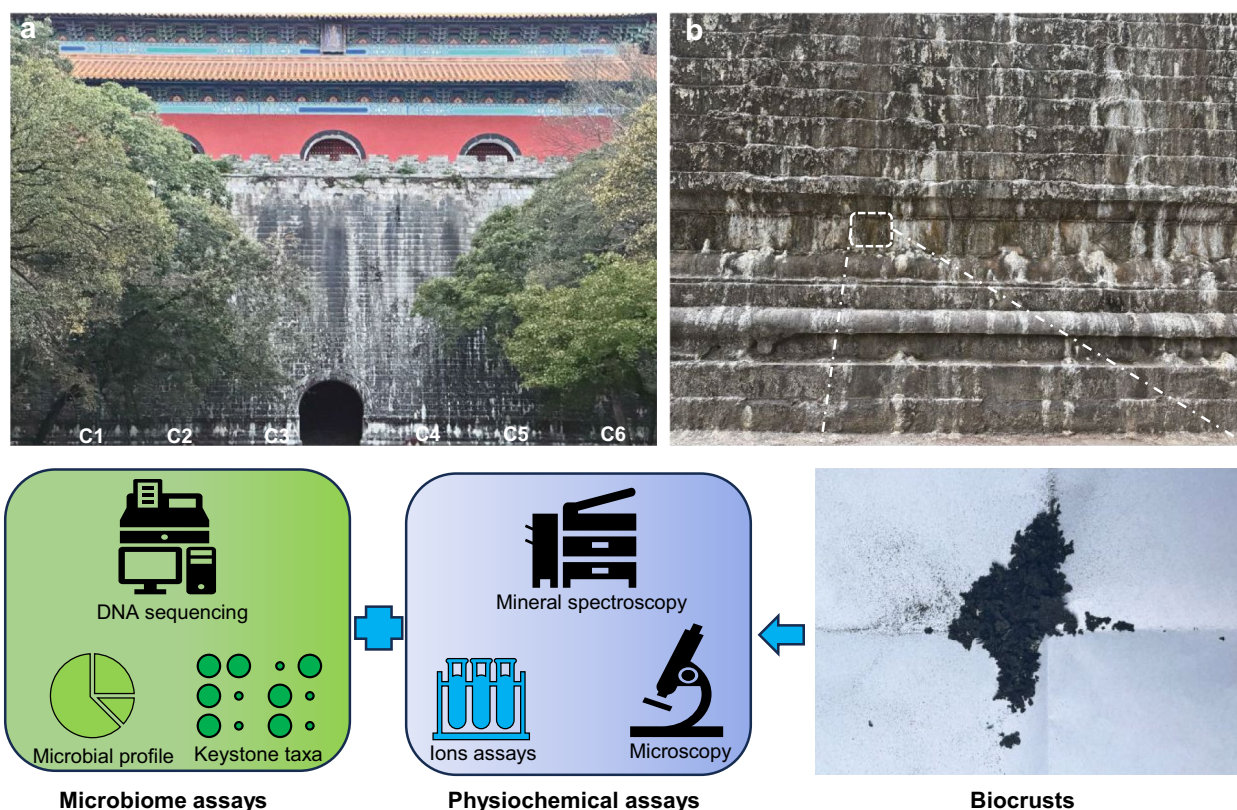


Fig. 1 An overview of the experimental design. **a** Panorama of the Rectangular Tower viewed from the front. **b** The limestone wall of the Rectangular Tower colonized by black biocrusts

the initial concentration would be diluted if necessary. Finally, the prepared supernatant was injected into an ion chromatograph to detect anion and cation concentrations. The anion concentration was measured using the Dionex™ Aquion™ ICS 2100 (Thermo Fisher Scientific, USA) system equipped with the column IonPac™ AS19 (4×250 mm). The cation concentration was examined using the Dionex™ ICS 6000 (Thermo Fisher Scientific, USA) system with the column IonPac™ CS12A (4×250 mm). The entire detection was carried out at 25 °C.

XRD analysis of biocrusts

The collected biocrust samples were dried and ground to approximately 300 mesh using a natural agate mortar. Before testing, the powdered samples were evenly spread on the sample stage and analyzed for phase identification using an X-ray diffractometer (D8 ADVANCE, Bruker, USA). The testing conditions were as follows: tube voltage, 40 kV; tube current, 40 mA; Cu target; scanning range, 5°–120°; step size, 0.0194°; scanning speed, 0.2°/min; and step-scan mode. The data were analyzed using Jade 9.0 software and the PDF-4–2009 database was referred for compositional identification.

Microbial activity analysis

To evaluate microbial activity of the collected biological crusts, microscopic analysis was conducted using the LIVE/DEAD™ BacLight™ Bacterial Viability Kits (Thermo Fisher Scientific, USA). Fresh biocrust samples were ground to 400-mesh using a natural agate mortar. A total of 3 mL of sterile physiological saline solution was added to 0.1 g of powder to make a suspension. Following the manufacturer's instructions, components A and B of the dye were mixed uniformly in equal volumes. Then, 3 µL of the dye compounds were thoroughly mixed with 1 mL of the suspension. The final mixture was stained in the dark at room temperature for 15 min. Eventually, 1 µL of the stained sample was loaded into a confocal dish and observed using an A1/SIM/STORM confocal microscope (Nikon, Japan).

Microscopy analysis

Before sampling, biocrusts were observed in situ with a portable microscope (3R-MSBTVTY, Anty, China) to obtain an overview of microbial community colonization. For scanning electron microscopy (SEM) analysis, the samples were dehydrated and dried in an oven at 37 °C for 1 h [14], and the observation surfaces were gold-coated under vacuum conditions. The morphological and structural characteristics of the prepared samples were

examined using the Quanta 250 (FEI, USA) scanning electron microscope.

DNA extraction and sequencing

Following the manufacturer's instructions, the DNeasy® PowerSoil® Pro Kit 50 (QIAGEN, Germany) was used to extract genomic DNA. The concentration and purity of DNA extracts were measured using a Thermo Scientific NanoDrop One (Thermo Fisher Scientific, MA, USA).

PCR amplification was performed on the V4-V5 regions of the ribosomal gene and the ITS1 region using the universal primers (bacteria: 515F-GTGC-CAGCMGCCGCGGTAA, 907R-CCGTC AATTC-MTTTTRAGTTT; archaea: Arch519F-CAGCCGCCGCGGTAA, Arch915R-GTGCTCCCCCGCCAATTCCT; and fungi: ITS1F-CTTGGTCATTTAGAGGAAGTAA, ITS2R-GCTGCGTTCTTCATCGATGC). Electrophoresis of PCR products is conducted on a 1.5% agarose gel to assess the fragment length and concentration. Samples with main bands falling within the normal range were suitable for further experimentation. Following concentration comparisons of PCR products using GeneTools Analysis Software (Version 4.03.05.0, SynGene), the required volumes for each sample were calculated based on equal mass, and the PCR products were then mixed. The PCR mixture was recovered using the E.Z.N.A.® Gel Extraction Kit (Omega, USA) D2500-03, with target DNA fragments eluted in TE buffer.

Library construction was carried out following the standard protocol of the ALFA-SEQ DNA Library Prep Kit, and the size of the library fragments was evaluated on the Qsep400 High-Throughput Nucleic Acid & Protein Analysis System (Hangzhou Houze Biotechnology Ltd., China). The library concentration was measured using a Qubit 4.0 (Thermo Fisher Scientific, USA). The constructed amplicon libraries were sequenced on either the Illumina or MGI platform using PE250 chemistry (Guangdong Magigene Biotechnology Ltd., China). The high-throughput sequencing data supporting this study were available in the National Center for Biotechnology Information (NCBI) under accession no. PRJNA1380739.

Bioinformatics analyses

Paired-end reads were assigned to samples based on their unique barcode and truncated by cutting off the barcode and primer sequences. Paired-end reads were merged using FLASH (<http://ccb.jhu.edu/software/FLASH/>) [15], a very fast and accurate analysis tool, which was designed to merge paired-end reads when at least some of the reads overlap the read generated from the opposite end of the same DNA fragment, and the splicing sequences were called raw tags.

Quality filtering of the raw tags was performed using fastp to obtain high-quality Clean Tags [16]. The tags

were compared with the reference databases (Silva database release 138.1, <https://www.arb-silva.de/> for 16S/18S; Unite database version 9, <https://unite.ut.ee/> for ITS) using the UCHIME Algorithm (http://www.drive5.com/usearch/manual/uchime_algo.html) to detect chimeric sequences, and the chimeric sequences were then removed [17]. For taxonomic assignment, a confidence threshold of 0.7 was applied for both databases. Finally, the effective tags were obtained. The entire microbiome analysis was performed using the Quantitative Insights into Microbial Ecology (QIIME, version 2021.4) tool.

Statistical analysis and data visualization

Violin plots were generated using GraphPad Prism 9.5 to display the physicochemical properties of the samples. Based on the QIIME 2 analysis, α -diversity box charts were plotted using GraphPad Prism 9.5. Bar charts illustrating the community structure of bacterial phyla across the six samples were created with GraphPad Prism 9.5. Heatmaps were performed to visualize microbial community structures at the bacterial genus level using the “pheatmap” package in R v.4.4.3 (<https://www.r-project.org/>).

Prior to network construction, Random Matrix Theory (RMT) was used to select an appropriate similarity threshold. Molecular Ecological Network Analysis (MENA) was conducted based on the relative abundances of OTUs across treatments. For network construction in this study, only OTUs that appeared in at least 4 of the 6 biological replicate samples were retained [18]. All network analyses were performed using the Molecular Ecological Network Analysis Pipeline (MENAP, <http://129.15.40.240/mena/>) [19]. The network was separated into different modules using the greedy modularity optimization method on MENAP. The resulting node and edge files were visualized in Gephi v.0.9.2. Based on differences in OTU relative abundance among modules within the co-occurrence network, both within-module connectivity (Z_i) and among-module connectivity (P_i) were calculated for each node. Mantel's test was conducted to examine relationships between chemical properties and keystone taxa using the “linkET” and “ggplot2” packages [20].

Results

Physicochemical properties of biological crusts suggest substantial calcium loss

The crystal structure and chemical composition of the biological crusts were analyzed by X-ray diffraction. The XRD patterns of all six samples exhibited major diffraction peaks at $2\theta = 20.5^\circ$, 26.3° , and 29.3° , which were identified as CaCO_3 , SiO_2 , and $\text{Na}_2\text{Mg}(\text{SO}_4)_2(\text{H}_2\text{O})_4$, respectively (Fig. 2a). Phase analysis indicated that the main components of the biological crusts were calcium

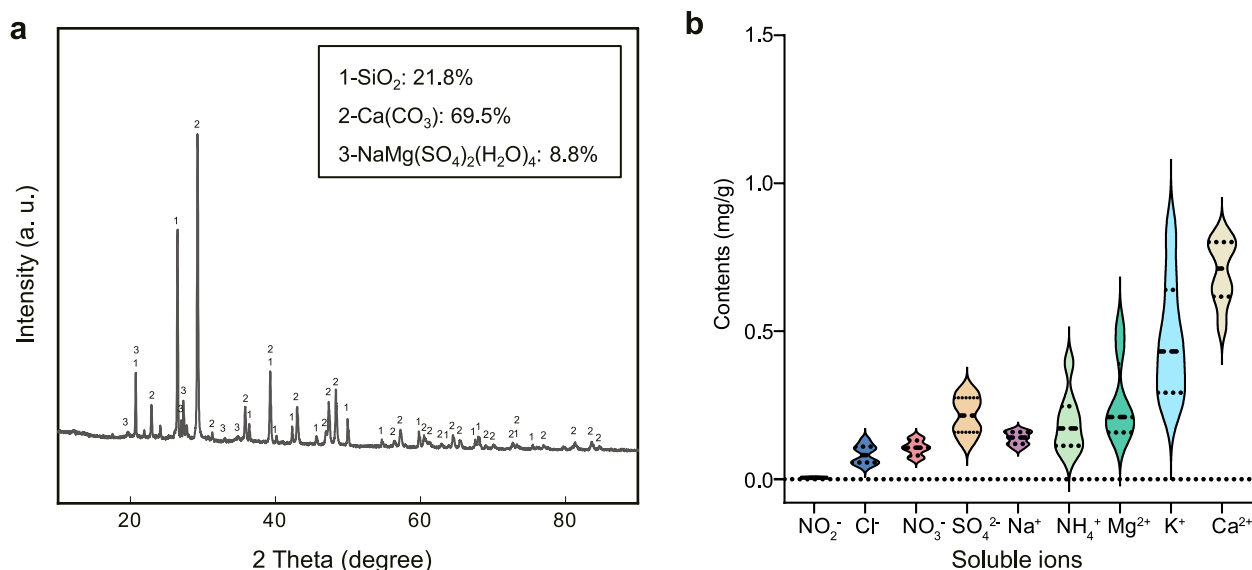


Fig. 2 Physicochemical properties of the biocrusts. **a** XRD analysis of the biocrusts. The digital values at the top of each peak correspond to the individual chemicals. **b** Contents of soluble ions of the biocrusts

carbonate, silica, and loweite, with consistent composition observed across all six samples. Previous studies report that the Rectangular Tower was built of limestone [21], which is composed primarily of calcium carbonate. This is also consistent with the XRD results in this study.

To investigate the potential chemical mechanisms underlying biocrust-induced biodeterioration, the soluble salt content of the biological crusts was measured. Four anions (Cl^- , SO_4^{2-} , NO_3^- , NO_2^-) and five cations (Mg^{2+} , NH_4^+ , Na^+ , K^+ , Ca^{2+}) were detected in the six samples (Fig. 2b). Overall, the anion content followed the order: $\text{SO}_4^{2-} > \text{NO}_3^- > \text{Cl}^- > \text{NO}_2^-$, while the contents of cations were ordered as $\text{Ca}^{2+} > \text{K}^+ > \text{Mg}^{2+} > \text{NH}_4^+ > \text{Na}^+$. Notably, Ca^{2+} content was consistently highest in the biocrusts, suggesting that substantial calcium accumulation. Taken together, it could be inferred that biocrusts have accumulated substantial amounts of soluble nitrate and sulfate, in the context of which the formation of biocrusts results in substantial accumulation of calcium in the Rectangular Tower's built environment.

Microbial interactions with stone induce the formation of biological crusts

Optical microscopy revealed an abundance of fibrous cells in the biocrusts at the site, indicating colonization by filamentous cells (Fig. 3a). Next, microbial activity in the biocrusts was evaluated using fluorescence staining to confirm the presence of a microbial community. Under the confocal microscope, the green fluorescent signals were significantly stronger in both quantity and intensity than the red fluorescent signals (Fig. 3b, 3c), indicating the presence of numerous metabolically active cells in the biocrusts [22].

To further confirm the morphology and microstructure of microbial communities in biocrusts, scanning electron microscopy (SEM) was performed at different magnifications. Similarly, hyphal cells were visible in the biocrusts (Fig. 3d), around which many microparticles were present (Fig. 3e). These microparticles might be fragments of deteriorated monuments or secondary biomineralization [23], mainly because the hyphae of fungi and cyanobacteria penetrate into the stone substrate, ultimately leading to the loosening and detachment of mineral particles [24]. Previous research has demonstrated that such (bio) mechanical damage is a key initial step in biodeterioration [1]. Furthermore, microbial extracellular polymeric substances, such as polysaccharides, lipids, and proteins, can exhibit chelating properties, thereby binding minerals effectively and accelerating biocrust formation [23, 25, 26].

These observations showed visual evidence of active microbial colonization within the biological crusts. Exploring the microbiomes in biological crusts will provide a crucial basis for understanding their role in the biodeterioration of stone monuments [27, 28]. Thus, further investigation into microbial communities and their interactions with the physicochemical properties of biological crusts is essential to uncover the microbiomes shaping the formation of biological crusts on limestone monuments in the Xiaoling Tomb of the Ming Dynasty.

Microbial diversity and community of biological crusts

Upon confirming the presence of a substantial number of active microorganisms, we investigated their diversity and composition in the biocrusts. This aimed to reveal the microbial communities associated with the

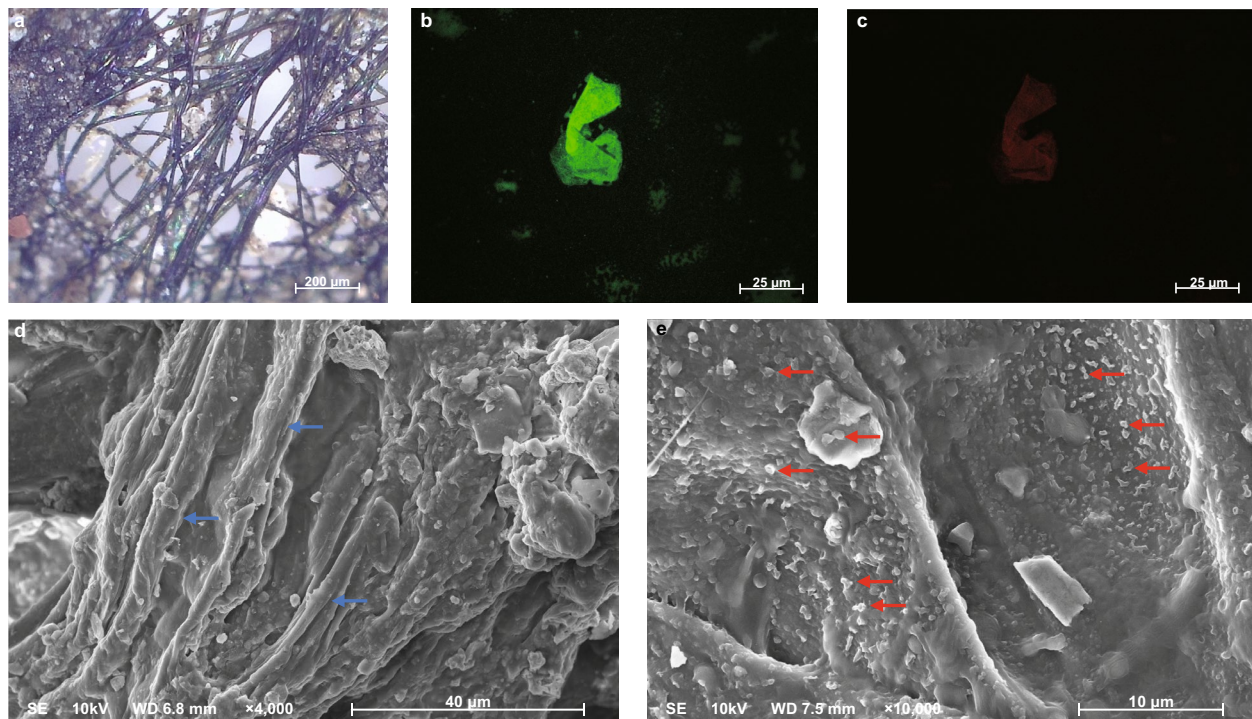


Fig. 3 Microscopic images of the biocrusts. **a** Portable microscopic image of the biocrusts in situ. Confocal microscopic images of the biocrusts after staining with green (**b**) and red (**c**) fluorescence. The SYTO9 green fluorescence nucleic acid stain can stain live cells, whereas the red fluorescence stain, propidium iodide, stains only dead cells. Thus, under confocal microscopy, green fluorescence indicates live cells, whereas red fluorescence represents dead cells. SEM images of the biocrusts at different magnifications **d** 4000 folds, **e** 10,000 folds). Blue arrows represent hyphal cells (**d**), whereas red arrows indicate mineral particles (**e**)

biodeterioration of the Rectangular Tower and to explore the potential mechanisms underlying biodeterioration driven by biocrust formation. Generally, bacteria showed the highest biodiversity, followed by fungi, while archaea had the lowest (Fig. 4a).

Bacterial communities across all samples were primarily composed of eight phyla, ranked in order of relative abundance as Cyanobacteria, Chloroflexi, Actinobacteria, Proteobacteria, Acidobacteria, Bacteroidota, Planctomycota, and Deinococcota (Fig. 4b). While Cyanobacteria are known for oxygenic photosynthesis, certain members of Chloroflexi are also capable of anoxygenic photosynthesis, suggesting that photosynthesis might be a crucial process involved in biocrust formation [29]. Consistently, at the genus level (Fig. 4c), the most abundant genera were members of the order Cyanobacteriales (accounting for approximately 19%), *Chalicogloea_CCALA_975* (approximately 13%), and *AKIW781* (approximately 10%), which were all members of Cyanobacteria.

Fungal communities in the biological crusts mainly consisted of Ascomycota, Basidiomycota, and Chytridiomycota (Fig. 4d). The dominant genera were *Ciboria* (accounting for approximately 12%), *Pleosporales_gen_Incertae_sedis* (approximately 11%), and *Anthracina* (approximately 7%, Fig. 4e). These fungi have often been observed in the arid and semi-arid environments [30],

which is consistent with the properties of the niches on the surface of stone monuments.

Interestingly, almost all the detected archaea belonged to the phylum Nitrososphaerota (Fig. 4f). At the genus level (Fig. 4g), the archaeal community was dominated by *Candidatus_Nitrocosmicus* and several genera within the family *Nitrososphaeraceae*. It is known that *Candidatus_Nitrocosmicus* is a classic ammonia-oxidizing archaeon capable of oxidizing ammonia to nitrite [31], while all cultured members of the *Nitrososphaeraceae* family are capable of chemolithoautotrophic growth through ammonia oxidation and carbon dioxide fixation [32]. Based on these metabolic characteristics, it can be reasonably inferred that the archaea detected in the biological crusts are primarily involved in the nitrogen cycle. This metabolic activity may also constitute a potential mechanism of biodeterioration.

Although we have gained a general understanding of microbiomes in biological crusts, the keystone taxa that shape their formation and drive the biodeterioration process remain unclear. Therefore, a combination of co-occurrence network analysis and Mantel tests could help identify the potential keystone taxa responsible for biocrust formation and the associated biodeterioration.

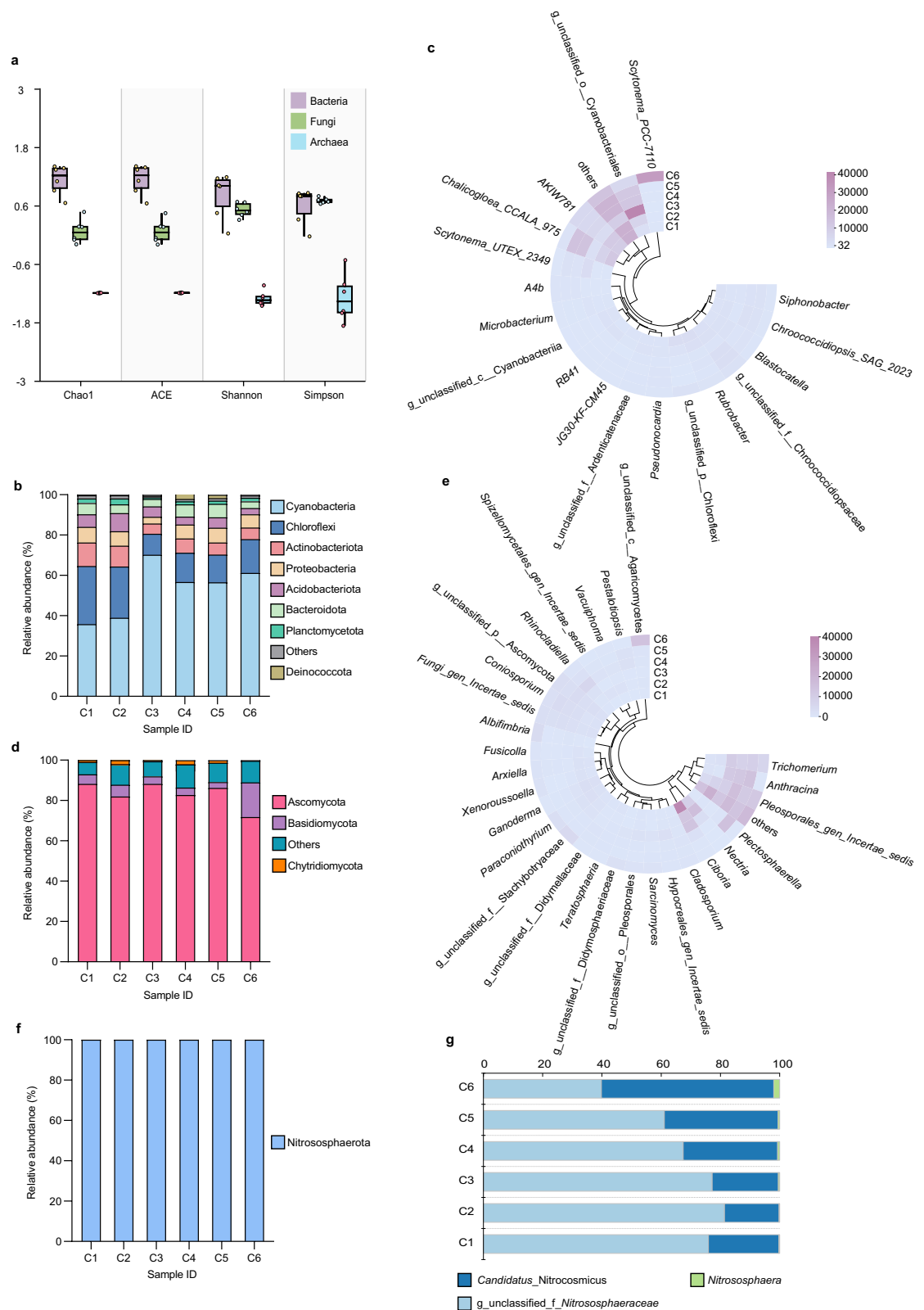


Fig. 4 Microbial profiles of the biocrusts. **a** Diversity indices of microbial communities in the biocrusts. **b** Bacterial populations at the phylum level. **c** Bacterial compositions at the genus level. **d** Fungal populations at the phylum level. **e** Fungal compositions at the genus level. **f** Archaeal populations at the phylum level. **g** Archaeal compositions at the genus level. Unclassified taxa and those with a relative abundance of < 1% were grouped as “Others”

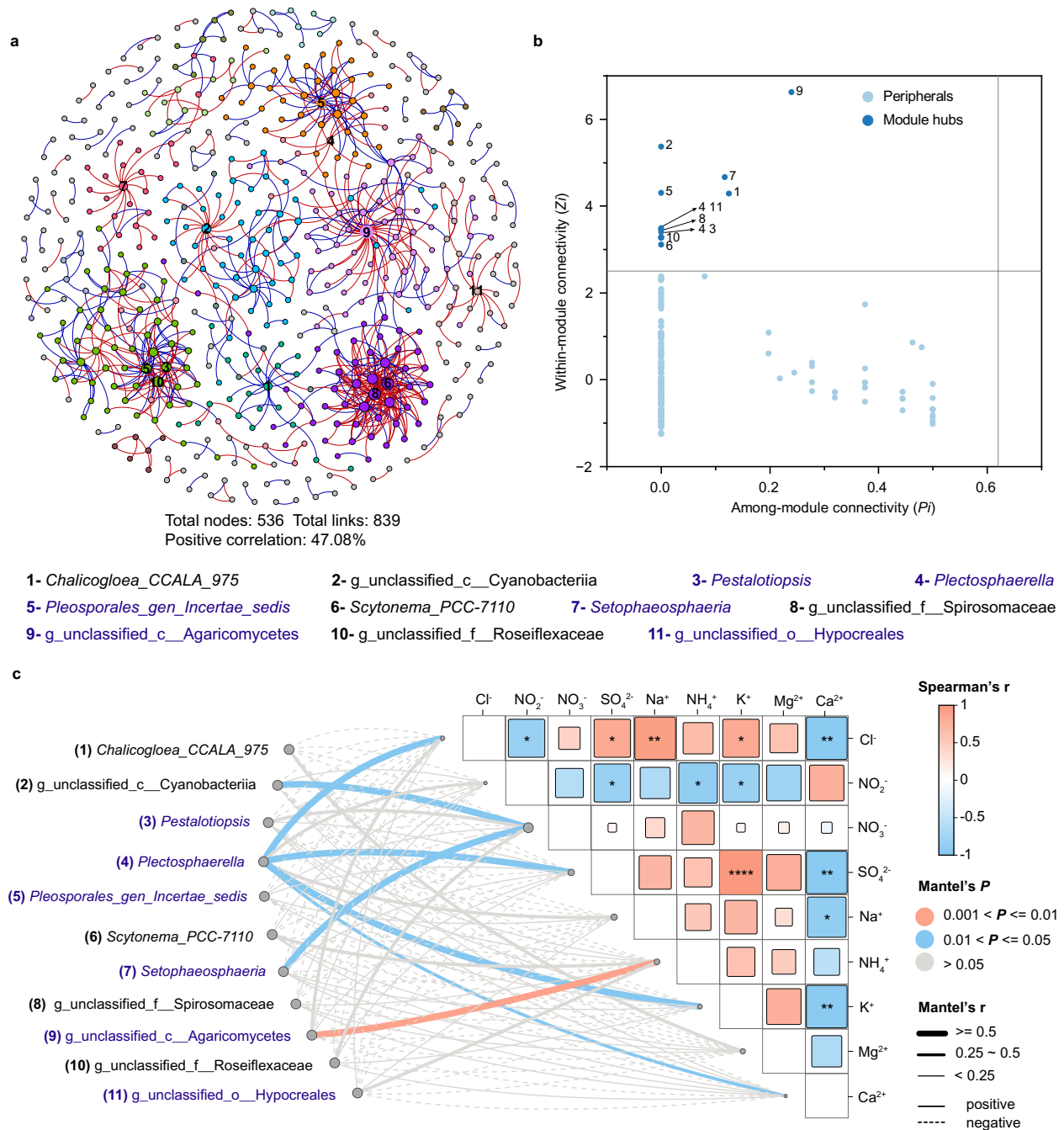


Fig. 5 Correlations among microbial communities and between keystone taxa and physicochemical factors. **a** Co-occurrence network among bacterial, fungal, and archaeal communities. Each node represents an OUT; node size indicates relative abundance; and node color is assigned by module, with modules with relative abundance less than 1% shown in gray. Red lines indicate positive correlations, while blue lines indicate negative correlations. **b** Module-connectivity of microbial communities. Each node represents an OUT and/or species from Panel **a** and is colored according to its topological role. The topological role of each species is determined based on a scatter plot of within-module connectivity (Z_i) versus among-module connectivity (P_i). **c** Correlation network between keystone taxa and physicochemical properties. Line thickness represents the strength of the correlation. P -values indicate significance levels (*, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; and ****, $P < 0.0001$)

Keystone taxa shaping the formation of biological crusts

To identify key species, we constructed a microbial co-occurrence network from bacterial, fungal, and archaeal communities (Fig. 5a). The network comprised 536 nodes and 839 edges, with 47.08% of the correlations being

positive. Based on differences in OUT relative abundance among modules within the co-occurrence network, the within-module connectivity (Z_i) and among-module connectivity (P_i) were calculated for each node. Species were categorized into the following four types: (i) peripherals

($Z_i \leq 2.5$, $P_i \leq 0.62$), which have few links and are almost always connected to nodes within their own modules; (ii) connectors ($Z_i \leq 2.5$, $P_i > 0.62$), which are highly connected to multiple modules; (iii) module hubs ($Z_i > 2.5$, $P_i \leq 0.62$), which are highly connected to many nodes within their module; and (iv) network hubs ($Z_i > 2.5$, $P_i > 0.62$), which act as module hubs and connectors. Except for peripheral nodes, the other three categories were considered keystone taxa due to their crucial roles in the network topology [19]. Thus, 12 keystone taxa belonging to 11 genera were identified (Fig. 5b) and highlighted in the network diagram (Fig. 5a). Moreover, most of these key species had high abundances within their own modules, and some (e.g., *Pestalotiopsis*, *Pleosporella*, *gen_Incertae_sedis*, and *Roseiflexaceae_unclassified* genera) were even clustered within the same module. Both fungal genera are typical saprophytes or facultative parasites, possessing robust lignocellulolytic enzyme systems. Positioned at the terminal stage of the organic carbon decomposition chain, they degrade recalcitrant organic matter, with *Pestalotiopsis* recognized as a symbiont in lichens [33]. In contrast, the bacterial genus in *Roseiflexaceae* may be at the front of the chain or work alongside other organisms within it. Powered by light energy, the microbiomes in the biocrusts could convert inorganic carbon into organic carbon or directly use early decomposition products, creating a small-scale carbon cycle nearby [34]. In addition, *Scytonema_PCC-7110* and *Spirosomaceae_unclassified* genera were located within the same module. *Scytonema* is among the most prevalent genera in biocrusts and plays a significant role in nitrogen fixation [35–37]. They are also producers of extracellular polymeric substances (EPSs), with their filaments encased in a thick, mucilaginous sheath composed mainly of polysaccharides and glycoproteins [38]. This sheath facilitates substrate attachment and promotes the formation of biocrusts, as shown in Fig. 3a. Meanwhile, *Spirosomaceae_unclassified* genera belong to the “cytrophagia” group and degrade complex polysaccharides [39]. In natural environments, the abundant EPSs produced by *Scytonema* serve as a crucial carbon and energy source for heterotrophic bacteria, including *Spirosomaceae_unclassified* genera. The latter, in turn, recycle and mineralize these organic compounds, thereby driving the carbon cycle. Thus, these two microbial taxa might form a consortium: *Scytonema* spp. provide *Spirosomaceae_unclassified* genera with organic carbon and nitrogen-fixation products, while the latter may help maintain local microenvironmental stability.

To better elucidate the relationship between key species and potential biodeterioration processes, we applied the Mantel test to compare the abundances of different key species with the physicochemical properties of the biological crusts. The Mantel test calculates the correlation

between two matrices and determines its statistical significance through permutation testing [40]. The Mantel test indicated that Ca^{2+} content was significantly correlated with SO_4^{2-} and Cl^- , but had no significant correlation with NO_2^- or NO_3^- (Fig. 5c). This finding diverges from our earlier inference that both nitrates and sulfates jointly promote biodeterioration, suggesting that the retention of Ca^{2+} in the bio-deteriorated crusts may rely more on the involvement of excess hydrochloric acid (or chlorides) and sulfuric acid to maintain its dissolution and migration. Meanwhile, the fact that SO_4^{2-} exhibits the highest concentration among all anions further supports this interpretation (Fig. 2b).

In the Mantel test, two genera, including an unclassified genus of the class Cyanobacteriia and *Setophaeosphaeria*, showed a significant correlation with NO_3^- ($P < 0.05$); and an unclassified genus of the class Agaricomycetes demonstrated a highly significant correlation with NH_4^+ ($P < 0.01$), suggesting that these taxa are likely involved in nitrogen cycling in the biocrusts [41–43]. Furthermore, the fungal genus *Plectosphaerella* showed significant correlations with SO_4^{2-} , Cl^- , K^+ , and Mg^{2+} ($P < 0.05$), suggesting that it may be a potential keystone taxon involved in biodeterioration of the stone monuments. Together, some genera in the classes Cyanobacteriia and Agaricomycetes, as well as *Setophaeosphaeria* and *Plectosphaerella*, are likely keystone taxa associated with biocrust formation and biodeterioration.

Discussion

To elucidate the association between microbial communities and biocrust formation on the limestone wall of the Rectangular Tower in the Imperial Xiaoling Mausoleum, this study integrated high-throughput sequencing data with physicochemical measurements. We examined not only bacteria but also profiled the archaeal and fungal communities simultaneously. By constructing topological networks and applying Mantel analysis, key microbial groups shaping biocrust formation were identified. Although 11 keystone taxa at the genus level were identified to shape the formation of biocrusts, statistical analysis indicated that only 4 of these keystone taxa (unclassified genera of the classes Cyanobacteriia and Agaricomycetes, as well as *Setophaeosphaeria* and *Plectosphaerella*) showed significant correlations with physicochemical factors associated with biodeterioration. This suggests that further research on the biodeterioration of stone monuments should extend beyond community diversity metrics and focus more on the functional analysis of microorganisms.

From the perspective of biocrust formation mechanisms, black biocrusts arise from synergistic interactions among multiple key microbial groups, with cyanobacteria and fungi playing essential roles in

structural development and functional maintenance. Previous studies have shown that, as early colonizers, cyanobacteria (e.g., *Scytonema* and *Chalicogloea_CCALA_975*) enhance their adhesion and expansion on mineral surfaces by secreting EPSs. EPSs not only trap dust particles and adsorb dissolved minerals (e.g., Ca^{2+}) but also promote the cementation of mineral particles and stabilize surface structures, thereby providing a critical physical foundation for biocrust formation [44, 45]. Meanwhile, cyanobacteria provide a continuous nutrient input through carbon and nitrogen fixation, and the organic matter they produce supports the growth of heterotrophic microorganisms, thereby driving local carbon and nitrogen cycling and maintaining microenvironmental stability [44, 46].

Previous studies have demonstrated that cyanobacteria in carbonate systems can simultaneously promote mineral precipitation and drive dissolution [44]. Many cyanobacterial species form biofilms on stone surfaces through EPS secretion, facilitating both CaCO_3 precipitation and micro-erosion. EPSs can capture Ca^{2+} and induce CaCO_3 reprecipitation, forming calcite crystals within pores that exert physical pressure and generate microcracks; concurrently, organic acids produced during cyanobacterial metabolism drive dissolution–precipitation cycles [23]. Coupled with environmental moisture fluctuations, these processes further increase porosity and weaken the stone structure. This dynamic transformation is largely regulated by water availability [27], which may explain insignificant correlations between cyanobacteria (e.g., *Chalicogloea_CCALA_975* and *Scytonema*) and Ca^{2+} observed in the Mantel test (Fig. 5c). In addition, some members of *Scytonema* (*Scytonema hofmanii*) produce secondary metabolites (e.g., scytonemin and cyanobacterin) that inhibit photosystem II [47], thereby suppressing the colonization of other photosynthetic eukaryotes [48]. This mechanism helps maintain a cyanobacteria-dominated community structure and may partly explain the relatively low fungal diversity observed in this study.

With further development of biocrusts, fungal taxa gradually colonize and play important roles in rock weathering. It has been reported that species of the genus *Plectosphaerella* are typically saprotrophic or facultatively parasitic, and their metabolic activities may regulate local microenvironments, thereby resulting in carbonate precipitation and reorganization and ultimately promoting biocrust formation [49]. Consistent with our findings, *Plectosphaerella* showed a significant correlation with Ca^{2+} , suggesting their involvement in calcium cycling, although the underlying mechanisms require further experimental validation. In addition, hyphae of *Pestalotiopsis* can penetrate microcracks in stone and generate periodic mechanical stress under wet-dry cycling, promoting crack propagation and structural

deterioration [50, 51]. In contrast, rock-inhabiting fungi (e.g., members of Agaricomycetes) are capable of both forming and dissolving calcium oxalate crystals. These fungi not only loosen mineral grains and induce exfoliation through hyphal penetration, but also promote mineral ion release via the secretion of organic acids and metal-chelating compounds [52]. Moreover, some fungal taxa (e.g., Dothideomycetes) can produce melanin, which absorbs strong radiation and alleviates UV stress, thereby creating a relatively stable microenvironment within the biocrust. This, in turn, facilitates the sustained development of black biocrusts and indirectly enhances their long-term erosive effects on the substrate [53, 54].

In this study, Ca^{2+} content was significantly correlated with SO_4^{2-} and Cl^- , but not with NO_2^- or NO_3^- , indicating that sulfate and chloride systems may dominate Ca^{2+} migration and retention in this environment. The highest concentration of SO_4^{2-} further supports its key role [55]. These observations have been validated in biocrusts at the Longmen Grottoes [56] and the Mayan Buildings [56]. Moreover, previous studies have shown that capillary action, evaporative concentration, atmospheric deposition, and salt weathering cycles all contribute to Ca^{2+} transport and redistribution [57]; meanwhile, EPSs can further influence Ca^{2+} retention through complexation and reprecipitation processes [23]. Therefore, the distribution and transformation of dissolved minerals are likely driven by a combination of biological and physical processes, a mechanism that requires further validation through in situ monitoring and uncolonized controls.

In conclusion, this study demonstrates that the formation of black biocrusts and the deterioration of stone materials are complex processes driven by the synergistic activities of multiple key microbial groups, particularly cyanobacteria and fungi. Given the limited correlations between keystone taxa and physicochemical factors and the strong environmental dependence of their functions, future studies should integrate multi-omics approaches, in situ monitoring, and stable isotope probing (DNA-SIP) to track active microorganisms and their metabolic processes, thereby providing a more robust theoretical basis for the conservation of stone cultural heritage.

Author contributions

X.L. and F.W. designed and managed the project. W.W., X.H. and X.L. performed laboratory assays, analyzed the data, and interpreted the results. W.W. wrote the manuscript. X.L. and F.W. edited and wrote sections of the manuscript. All authors reviewed the manuscript.

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Data availability

The raw gene sequence data supporting this study were available in the National Center for Biotechnology Information (NCBI) under accession no. PRJNA1380739.

Declarations

Competing interests

Xiaobo Liu is an Associate Editor for Environmental Microbiome, but was not involved in the peer review of this manuscript in any way. The authors have no other competing interests.

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